Giovanni Toto

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About Me

Date of birth: May 12, 1997
Place of birth: Padova, Italy
Nationality: Italian

Research Interests

- Bayesian nonparametrics
- Density estimation
- Factor modeling
- Functional data analysis
- Random partition models

Current Position

Oct. 2025 - ongoing
Postdoctoral associate
Department of Statistics and Data Sciences, The University of Texas at Austin
Supervisor: Yang Ni

Education

Oct. 2022 - Sept. 2025
PhD in Statistics
Department of Statistical Sciences, University of Padova
Thesis title: Bayesian Nonparametric Multiple Random Partition Models
Supervisor: Antonio Canale
Co-supervisors: Peter Müller, Abhra Sarkar

Oct. 2019 - March 2022
Master's degree in Statistical Sciences
Department of Statistical Sciences, University of Padova
Thesis title: A Topic Modeling Algorithm for Microblogs
Supervisor: Emanuele Di Buccio
Final mark: 110/110 cum laude

Oct. 2016 - Nov. 2019
Bachelor's degree in Statistics for Technology and Science
Department of Statistical Sciences, University of Padova
Thesis title: Design and Implementation of a Video Search Engine and a Graphical User Interface for Sentiment Analysis
Supervisors: Massimo Melucci, Emanuele Di Buccio
Final mark: 101/110

Visiting periods

March 2024 - Sept. 2025
Department of Statistics and Data Sciences, The University of Texas at Austin
Supervisors: Peter Müller, Abhra Sarkar

Research Activities

June 2022 - Oct. 2022	Research fellowship Department of Statistical Sciences, University of Padova Title: Development of R libraries for the implementation of Bayesian models for sc-RNA-seq data Supervisors: Antonio Canale, Davide Risso Implementation in R and C++ of Bayesian models for sc-RNA-seq data.
March 2020 - April 2020	Research fellowship Department of Information Engineering, University of Padova Title: Design, implementation, and use of a computer system for online acquisition of multimodal documents and real-time opinion analysis Supervisor: Massimo Melucci Design and implementation of a graphical interface allowing real-time analysis of opinions expressed in a video through multimodal sentiment analysis.

Publications

Journal

- Canale, A., Galtarossa, L., Risso, D., Schiavon, L., **Toto, G.** (2026). Structured factorization for single-cell gene expression data. *Journal of the Royal Statistical Society Series C: Applied Statistics*

Conference proceedings

- **Toto, G.**, Di Buccio, E. (2022). A modular approach to topic modeling for heterogeneous documents. *Proceedings of the 12th Italian Information Retrieval Workshop 2022, Milan, Italy, June 29-30, 2022*. CEUR-WS.org
- Li, Q., Stefani, A., **Toto, G.**, Di Buccio, E., Melucci, M. (2020). Towards multimodal sentiment analysis inspired by the quantum theoretical framework. *3rd IEEE Conference on Multimedia Information Processing and Retrieval, MIPR 2020, Shenzhen, China, August 6-8, 2020*, 177–180

Preprints

- **Toto, G.**, Canale, A. (202+). Bayesian local clustering of functional data via semi-Markovian random partitions
- **Toto, G.**, Müller, P., Sarkar, A. (202+). Bayesian semiparametric multivariate density regression with coordinate-wise predictor selection

Presentations

Seminars

Oct. 2026	Condition matrix completion for polypharmacy medication information The University of Texas at Austin, Austin, TX
Sept. 2026	Condition matrix completion for polypharmacy medication information University of Colorado Anschutz Medical Campus, Aurora, CO

Conference talks

July 2026	Condition matrix completion for polypharmacy medication information ISBA 2026, Nagoya, Japan
June 2022	A modular approach to topic modeling for heterogeneous documents IIR 2022, Milan, Italy

Posters

June 2026	Condition matrix completion for polypharmacy medication information BAYSM 2026, Chiba, Japan
April 2026	Bayesian multivariate density regression with coordinate-wise predictor selection COTS, Austin, TX, USA
Sept. 2025	Bayesian multivariate density regression with coordinate-wise predictor selection SISBayes, Padova, Italy
June 2025	Bayesian multivariate density regression with coordinate-wise predictor selection BNP 14, Los Angeles, CA, USA
April 2025	Bayesian multivariate density regression with coordinate-wise predictor selection BOSS 2025, College Station, TX, USA
July 2024	Local Bayesian clustering for functional data ISBA 2024, Venice, Italy
June 2024	Local Bayesian clustering for functional data BAYSM 2024, Venice, Italy

Awards

- Best poster award at BAYSM 2026, Chiba, Japan (June 2026).
- Travel award at BNP 14, Los Angeles, CA, USA (June 2025).
- Best poster award at ISBA 2024, Venice, Italy (July 2024).